"You Creep! It Really Worked!": An Empirical Study of Telephone Scams with Cloned Familiar Voices and Trusted Caller IDs

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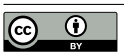
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Abstract

Telephone scams often attempt to defraud or steal the identity of individuals by eliciting a response to a pressing request for payments or submission of information such as social security numbers. These calls usually come from numbers that individuals have not encountered before, and the request is passed through as a pre-recorded message with a generic voice. But AI-enabled voice cloning and the ability to spoof Caller IDs have given scammers the opportunity to run schemes targeted individuals with a cloned familiar voice coming from trusted numbers. This paper reports the findings from an empirical study that replicates this scenario with 14 participants (7 pairs of family relatives of friends) to capture the experiences of receiving such a scam call and responses to it. The results of our thematic analysis show that the familiar voice, coming from a trusted number of a family relative or a friend, is highly persuasive towards deceiving the call receiver (i.e., *callee*) to indeed follow through with the scammer's request. The *callee*, together with the *caller* or the participants who volunteered their voices for cloning, saw this scam working particularly in the context of family emergencies, as real-world reports have surfaced about children held for ransom, grandchildren under arrest, or relatives in car accidents. The *callers* and *callees* saw no immediate way to fend off these "*family emergency scams*" than for families and friends to work on "family/friend codewords" that are hard to be inferred by the scammers (and thus cloned). We discuss our findings towards the development of user-centered interventions that would facilitate the detection of a wide range of AI-enabled voice cloning scams, in addition to the suggested personal ways of scam detection.

CCS Concepts

• **Security and privacy → Usability in security and privacy**; **Social aspects of security and privacy**; **Social engineering attacks**.



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Keywords

AI-enabled voice cloning, telephone scams, familiar voice, Caller ID spoofing, contextual scam warnings

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1 Introduction

Telephone scams, robocalls, or any form of unwanted calls from spoofed Caller IDs is an inconvenience that individuals, despite aggressive regulatory and technical protections [16, 25], have not yet found a way to avoid [14] (we speak from the US context of the study, but scam calls are regular bother for people outside the US too [32]). Most of these calls attempt to "scam" people, i.e., defraud them financially or steal their identifying information, such as their Social Security number [68]. The nature of the telephone scam varies, but often involves known and proven pretexts such as impostor scams on behalf of government tax of social security agencies, investment opportunity scams, health insurance scams, lottery scams, and job or money-making scams [23].

The FTC reported that there were 857,346 reports of telephone scams in 2024 for a total of $2.9 billion in losses and in the first quarter of 2025, there has been 217,814 reports for a total of $781.9 million in losses [28]. A new type of telephone scam has recently emerged – a *family emergency scheme* where the *caller*, sounding very much like a family member, pretends to be in trouble and then asks for the *callee* to send money as a help [63]. The difference in this new pretext is that the voice on the other side of the call is *cloned* using readily available generative AI technology, making it indistinguishable from one familiar to the *callee*, thereby increasing the probability that the *callee* will fall for the scam [33].

The family emergency schemes place the *callee* a victim under an *economic harm*, but also *harms to mind* as the *callee* is tricked to commit an action, possibly inducing a state of emotional distress. Unlike the other unwanted calls, this new type of *AI-enabled voice cloning* scam also makes the original voice owner the subject of *impersonation* harm, as their voice has been used without consent for exploitative purposes [33, 38]. Several incidents demonstrate the particular deceptiveness of this scheme practice – a mother

getting a call from a kidnapped daughter [65], grandparents getting a call from a grandson under arrest [69], or family members being called to help from a relative involved in car accidents [62].

Past work on telephone scams shows that they: (1) come from spoofed Caller ID (formatted such that it matches the first three or six digits of the recipient's phone in order to entice them to answer the phone call believing it is a neighbor or local organization, e.g., a school); (2) are automated (in the sense of using an automated calling logic targeting large set of numbers); (3) and usually placed as "robocalls" (i.e., with a pre-recorded message, by an actual person or, more recently, by an AI-generated generic voice) [55, 56, 67, 68]. Authentication solutions such as STIR/SHAKEN have been introduced to help users "verify numbers in order to confirm that a call is coming from the source it claims, but neither this nor the introduction of the national Do Not Call Registry," where individuals can enter their number to avoid unsolicited calls, has proven effective in deterring scammers [16, 24]. Anti-spam solutions, such as "anti-robocall" apps [49, 61], may aid users in fending off known robocalls, but they cannot anticipate a potential scams and might preclude a *callee* from receiving a wanted call, such as doctor appointment reminders, prescriptions, or similar [16].

As spoofing a Caller ID is trivial [55], past work has shown imposter scams work, and scammers could effectively retrieve individuals' Social Security numbers by using a local Caller ID trusted by the victim [67]. But how would individuals respond if a call: (1) comes from a (spoofed) Caller ID that matches the number of a family member or a friend, matching all ten digits; (2) is manually crafted (targeting a *calle*e number that has the family member's or friend's name in their contacts with high probability); (3) and the pre-recorded message is an AI voice clone of the family member's or friend's voice? In this paper, we set to investigate, with an Institutional Review Board (IRB)-approved non-full disclosure protocol, a close resemblance of this situation and gather the first-hand impressions of both the *callee* and *caller* about the deceptive nature of the *family emergency scheme*. Given that the aforementioned harms are non-trivial, we crafted a non-full disclosure research protocol that used a predefined message in a couple of pretext variants: (i) loss of shared membership; and (2) identity surfaced on the dark web. We avoided using an actual emergency as a pretext in the messages to prevent causing harm, and the voice clone recording was crafted politely to avoid conveying urgency or pressure in the request. (see Appendix 4 for more details on the study procedure).

We recruited seven participants who had already explored cloning their voice themselves unrelated to the study with solutions like ElevenLabs [17]. First, we obtained consent from participants (hereafter: *caller*) to clone their voice using an education-licensed ElevenLabs account (recording downloaded and immediately deleted from the account) and select a family relative or a friend (hereafter: *callee*). Instead of cold-calling the *callee* directly with the voice-cloned recording of one of the pretexts, we instead asked the *caller* to call the *callee*, put them on speakerphone (so we do not see their number, name, and any private information) so we can offer them the option to participate in the study and agree to listen to a pre-recorded message. On agreement (if they disagreed, the *caller* would have thanked them, closed the call, and worked to select another close person who might be open to participate). Once the pre-recorded message was played (a recording of one of the aforementioned pretexts with the cloned voice of the *caller*), the researchers asked the *callee* about the authenticity of the voice they heard as well as the plausibility of the call context and the request. Following that, we offered a full disclosure through debriefing about the nature of the study and proceeded with the remaining of the interview with both the *caller* and the *callee*.

Findings: The *callees*, privy of traditional robocalls and telephone scams in the past (e.g., imposter scams, insurance scams), found the cloned familiar voice in the pre-recorded message, taken together with the fact that it comes from a known Caller ID, a bit "***creepy***" [**Theme I**]. This feeling of creepiness was expressed in the scientific sense where one enters a state of reservation due to ambiguity whether there is something to fear and/or ambiguity about the very nature of the threat [44] (not the colloquial meaning of "creepy" as in "nervous" or "frightened"). The ambiguity of this unusual situation made the *callees* share that they would, if they receive such a call for real (and not as a part of a study), not hesitate to follow up with what is requested of them. The *callees*, together with the *callers*, shared that they would particularly see such a scheme of familiar voice cloned over the phone working well in the context of "***emergency***" [**Theme II**] (deliberately avoided as a safeguard for participation). As this scheme is relatively new and fast developing, our participants saw the only way to protect themselves is through the use of "***family/friend code words***" [**Theme III**], a secret piece of information known only among the family members, relatives, or friends that is hard to be found out by scammers and cloned.

Scope and contributions of this work:

- We offer evidence of first-hand lived experiences with AI-enabled voice-cloned telephone scams that exploit familiar voices and are situated in realistic settings;
- We describe a systematic approach for an empirical test of the deception potential of AI-enabled voice-cloned telephone scams using a family emergency scheme pretext;
- We provide insights about personal protection strategies against familiar cloned voices and trusted Caller ID telephone scam schemes;
- We discuss technical protection strategies from a systematic, usable security perspective as a scalable response to emerging types of AI-enabled voice cloning telephone scams.

2 Background/Related Work

2.1 AI-Enabled Voice Cloning

AI-enabled voice cloning and the close class of "audio deepfakes" (here, the source of the voice does not have to be an actual human, that is, the speech could simply be synthesized) are particularly interesting forms of deception as individuals need to rely on potential familiarity with the speaker's voice or the context in which they could be interpreted as a legitimate or fake (compared, for example, with video deepfakes or AI-enabled video cloning [7]). During the primaries, the familiar voice of then President Joseph Biden was impersonated in an AI-enabled voice-cloned message that was circulated through telephone scams urging people across New England not to vote [70]. A voice cloned message of the LastPass CEO was left on the phone of a LastPass employee with a spoofed Caller ID asking for sensitive documents to be forwarded

to given receiver [37]. And a voice cloned message of a UK-based company CEO persuaded an employee to transfer almost a quarter of a million pounds to a scammer-controlled account [11]. In all cases, the familiarity of the speaker was the key element of persuasion that the scammers bet on, praying on the convincing effect when delivered through a phone call or a voice message.

The reason the scam through a familiar voice is that humans perform very poorly in discerning a legitimate voice from a cloned one. Studies using known datasets such as the ASVspoof2019 and ASVspoof2021 [37, 50] have shown that humans could perform as best as 73 to 80% accurate in spotting cloned voices [41, 48]. Another more realistic evaluation brought this number to 50.57%, with samples generated using with more advanced speech synthesized in the datasets for testing [41]. It must be noted that participants in these studies evaluated the speech only on the subjective impression of the "naturalness" of the speech without any context of the recorded messages. Using ElevenLabs or Wavefake as voice generators, the human performance for detection peaked between 60.8 and 63% [6, 72]. Even people with visual disabilities, who are exposed to synthesized speech more than their sighted counterparts, could not perform better than 54% accuracy for ElevenLabs cloned speech with a convincing context, resembling the ones used in the telephone scam schemes [60].

2.2 Telephone Scams

Unwanted calls come in many forms and shapes, often automated (or "robocalls"), sometimes as *spam* (e.g., telemarketing) but much more as *scams*, or ones aiming to exhort money or compromise the recipient's private information. As call verification is a not a unified feature among telephone providers [16], telephone scams come from a spoofed caller ID and defraud individuals or steal their identitiesy to an alarming degree [22]. Individuals could use blocking apps [26] or screen calls on their voice mails [59], but telephone scammers are persistent and, finding a way to evade the blocking, tend to "pressure" recipients with repeated voice mail spam in order for them to pick up the phone [55].

Telephone scammers know that individuals do not have much time to assess the call for authenticity and context plausibility, compared to, for example, with an SMS text or an email phishing request. Even if individuals could see scam warnings about the unwanted calls – for example a "Scam Likely" tag on the calling screen [45] (akin to the phishing warning for emails and SMS texts [30, 46]) – these only tell a *callee* that the phone number could not be verified; it doesn't tell anything about the content of the scam itself (that, could be cloned familiar voice). The time limitation puts individuals in a position to pick up a call even if they might not trust the *caller* (e.g., it comes from an unknown number or is labeled as "Scam Likely"), as they might feel they have to answer it anyhow because it might be related to work, a personal engagement, or an expected callback. As scammers rarely reuse phone numbers [56], individuals have mostly themselves to rely on for avoiding telephone scams.

But, extensive experience in fending off scams could be only helpful for oneself to recognize imposter, investment, money-making, or insurance scams. Semantically, these scams share common persuasive elements that pray on the vulnerable positions of the *callees*, usually associated with scarcity in time, finance, or benefits and combined with authority (e.g., IRS or a bank) [56]. The new type of scams adds a persuasive element that prays on emergency, but with a *twist*: the voice is not just a random speaker or a generic one that comes in available voice synthetization tools, but a clone of familiar one from a relative or friend [63]. Dubbed as *family emergency schemes* or *grandparents scams*, this new type threatens not just to subject the *callees* to economic harms (as the familiar voice requests money for being held against their will), but also harms of mind (emotional distress), and, *callers* a subject to impersonation harm (use of their voice without consent and knowledge) [33].

3 Paradigm

The *family emergency schemes*, facilitated by *voice cloning*, introduce a new security paradigm as they enable an unprecedented "attacker" advantage – *freely* available voice cloning software and source of target voice sampling online. This advantage, at present, is matched with no "defender" options (warnings, telephone network blocking/filtering, etc.) at the disposal of targets – the individual who's voice is cloned and the individual that cloned voice is used to deceive. The disadvantage is further exacerbated by three additional conditions: (1) any implicit *trust* built over the years, if not decades, in the telephone networks to deliver authentic voice calls is suddenly questioned ("*Is the voice on the other end, I am familiar with, actually coming from the person I know and their number I saw on the screen that just called me?*"); (2) a widely *available voice samples* online for target cloning that an attacker could abuse (e.g., social media posts, YouTube videos, even an unsuspecting telephone call to sample one's speech by recording it for "quality assurance purposes"); and (3) a *constantly improving voice cloning technology* that, for now, allows indistinguishable cloned voices to be *played* as recordings (unidirectional), but in the near future could well allow cloned voices to *interact* with the targets (bidirectional) [34].

The loss of implicit trust in telephone communication, or any voice communication for that matter, is *irreversible* because from now on, any call could be made by a cloned voice and the *callee* could not be 100 percent sure they speak with the real authentic person behind the voice on the other hand. But there is also another associated *irreversible* restructuring of interpersonal trust because from now on, any familiar contexts for emergency requests might be plausible and the *callee* could not be 100 percent sure that these are real or not. The spoofing of the CallerID appends this loss of trust, in addition, because combined with the convincing voice and a plausible context, it does make a compelling case *not to* simply dismiss the call as just another scam.

While it is tempting to classify the *family emergency schemes* facilitated by *voice cloning* as just another scam because the goal is the same (defraud or identity theft) or because they rely on the persuasion principle of urgency/scarcity, the premise of impersonation and the lack of technical or usable security "defense" is rather new. Other scams usually impersonate third parties (impostor, investment opportunity, health insurance, lottery, job scams) or unknown individuals (romance scams). But the *family emergency* scam impersonates a *first party, known individual* that the target most probably *trusts*, and, is in a more than just a transactional relationship with. An added dimension here is that the target not just feels the urgency/scarcity to respond but also a *responsibility* to do so even if

the presence of doubts (imperfect cloning, not entirely plausible context or so), an argument absent from any scam, phishing, or traditional social engineering deception so far.

For example, in impostor, romance, job, or lottery scams the "stakes" of too good (or dire) to be true rejection of a request due to being skeptical might be inconsequential (the government won't go anywhere, another romance might line up, a new job opportunity too, as well another lottery chance is always there). In contrast, the common sense skepticism as a "defense" might turn out very consequential against a *family emergency schemes* (a suffering of a close family friend, immediate harm) and rather costly as it ultimately would result in general loss of trust. And, unlike in any of the other aforementioned scams, in this one, only, there are actually *two*, instead of *one* targets that are exposed to harm – the *callee* and the familiar person whose voice is cloned.

So, any usable security protection, currently absent, exposes the need to protect both of these target individuals in the context of "defending" *family emergency schemes* facilitated by *voice cloning*. If the individuals could reinforce their skepticism by seeing a "Scam Likely" label on their screen, no such option exists here as the call comes from a familiar Caller ID and the known voice. The plausibility of the scam request is the only element where the *callee* could fend off the scam call by dismissing it, as our study indicated, based on a shared secret or a "family/friend codeword" but this is a protection independent of any technological intervention. What is more, the target individual whose voice is cloned has no way of preventing the malicious use of their voice or speech found online or obtained directly from them. It seems infeasible to resort to extreme measures of removing one's online voice footprint or asking anyone within a family or friend circle to develop a secret codeword. The paradigm of irreversibly undermining the societal *trust* in a system of voice communication by the indistinguishability of fabricated voices (and soon entire reasoning personas) thus also challenges the paradigm of centering security interventions around a *single* user assumed to be a target in a *one-to-one* exploit.

A shift to *one-to-many* "defenses" is inevitable, considering that the 'many' (in this case 'two') are *related* victims in a dangerously deceptive social engineering attack. The consideration entails an obvious broadening the "Scam Likely" warnings to include "Voice Clone Likely" type of alerting, but also necessitates an element of *verification* that has to eventually append, if not entirely eliminate, the need for mutual "codeword" through technical means. This design consideration, from a *one-to-many* kind of exploit, hopefully, would also open the possibility to expand and incorporate a user-centered alerting in where conversational voice clones will eventually dominate the *family emergency schemes*. Also, the consideration equally pertains to the already use of "face swapping" technologies in multimedia variants of the traditional version of the romance scams, that no doubt will be utilized in future *family emergency schemes* [9].

4 Methodology

4.1 Recruitment and Sampling

We obtained an Institutional Review Board (IRB) approval for a non-full disclosure study with adults from the US who have independently explored cloning their voice themselves with solutions like ElevenLabs [17]. We used a random sampling procedure through snowballing participants from a personal pool (*callers*) who were free to select a choice of *callee* in consultation with us (after obtaining consent and informing them about the details of the study). The overall interview took 60 minutes, took place on Zoom, was audio-recorded only, and the voice messages were played through speakers. The compensation for the *caller* was $60 Amazon eGift card for the *caller* and a $40 Amazon eGift card for the *callee* (the eGift card number and code were read out loud at the end of the interview after the recording was stopped so the participants could redeem it at the conclusion of participation on their own, or, an Amazon eGift card was sent to a voluntarily provided email for delivery that was not recorded, collected, or included in any way in the analysis of the results to preserve anonymity). We opted for free and available solution as past voice cloning scams have been aided by ElevenLabs (notably faking President Joe Biden's voice in an early electoral disinformation campaign [70]) and as a demonstration of a threat model where an attacker need not be versed in voice cloning but could well opt out to simply sample someone's voice (from online recordings, posting on social media, etc.) and generate a clone in few clicks [17]. Another reason was to ensure that the participants were familiar with the voice cloning procedure and had already accepted the terms of use independently from our study for the cloning service of ElevenLabs [18].

We did not use the concept of *saturation* for determining our sample size [29] following criticism of applying this concept to non-grounded theory qualitative research [8, 53]. Instead, we turn to the concept of *information power* as proposed by Malterud et al. [42], which allows evaluating our sample size along a five-axis model: *aim*, *specificity*, *use of theory*, *dialog*, and *analysis*. The *aim* of our study was at the narrower side of the spectrum, given that prior research has explored general telephone scams but not ones with AI-enabled cloned voices using freely available generative AI tools nor family-related scams situated in realistic conditions [16, 49, 59, 67, 68]. The *specificity* of our study is dense, as all participants were technologically literate, experienced various types of scams over the phone or emails/SMS texts, and were ideally positioned to share lived experiences with telephone calls that might invoke a family request pretext, that hypothetically, might be misused by adapting to emergency towards financial fraud or identity theft.

Our final analysis (thematic, per [8]) relies on very little pre-existing *theory*, i.e., it is more exploratory. The strength of the *dialog* of our interviews is strong. The involved researchers had extensive experience conducting interviews in realistic settings that involve non-full disclosure protocols and protocols with deception about cybersecurity problems (within the ethical limits for participation per the Menlo/Belmont reports [13, 66]). Finally, our *analysis* was conducted across the cases, as our goal was to capture diverse yet subjective first-hand responses and experiences under realistic conditions involving an emerging type of telephone scam—one that, fortunately, few people have been subject to so far (and we hope it stays that way or, better yet, disappears entirely). For a catalog of AI-enabled cloning incidents, see [3, 5, 52]). Given these factors, we concluded our data collection with seven pairs of *callers* and *callees* or 14 participants in total, deeming our study's *information power* sufficiently strong.

4.2 Research Procedure

The study procedure is shown in Figure 1. We forwarded a recruitment email to potential participants (see Appendix A), including a detailed information sheet about our study, a Calendly link to schedule a time for the interview, and details for the Zoom call. Once the *caller* joined the Zoom call at their chosen timeslot, the researcher worked with them to choose a *callee* (a close person familiar with the *caller's* voice). Here, we informed the *caller* about the potential of however minimal, but still valid and possible, harms of mind [33] and advised them to select a *callee* they believe has the least probability to experience any adverse effects from participation (we also ensured this during obtaining the consent form the *callee*, but we wanted to make double sure our tests do not cause more than minimal harm). The *caller* was also informed that the study is anonymous – confidential while the study takes place, but the answers given in the audio-only recording of the study are anonymized for later reporting in a scientific paper (Zoom generates a transcript upon the completion of the interview, we immediately downloaded it and anonymized it, and immediately delete the audio recording afterward).

Once on the call, we shared the Rainbow Passage with the *caller* as a standardized paragraph utilized for eliciting speech samples [19] (also used in ASVSpoof2019 and ASVSpoof2021 voice clone challenges [40, 50]). To create the voice sample, we muted ourselves, restarted the Zoom audio recording, and asked the *caller* to read the Rainbow Passage contents. Once we were done, we stopped the Zoom audio recording, produced an audio file, uploaded it to ElevenLabs, and immediately deleted the audio and the transcribed Rainbow Passage from the Zoom servers. Once we created a temporary clone of their voice, we then worked with the *caller* to select one of the two pretext options, either for confirming a shared account or for a potential identity record on the dark web. We selected these two pretexts to be believable and plausible for the *callee*, yet not to be sounding emerging. We also selected them to correspond to the sample telephone scams offered by the FTC for demonstration purposes of the most prevalent ones [25].

The "pretext A" was initially designed to be about a shared Amazon Prime membership, but we allowed the *caller* to adapt it to any shared membership they have with the *callee*. The "pretext B" scenario operated on the assumption that the *callee* knew the *caller*'s address, but we permitted minor modifications, such as confirming a phone number or date of birth, if the *caller* felt it was more probable that the *callee* would know that information about them. Here, even though the pretexts asked for inputs like the "four digits of the membership credit card (pretext A)" or "address, phone, date of birth (pretext B)" we never actually asked the *callee* exactly about them (no personal information was disclosed), but asked only if, assuming they accepted the call as plausible, they would have hypothetically disclosed to what would have been a real scammer on the other side of the call.

The selected and adapted pretext was then entered as a text in ElevenLabs, customized with pauses, breathing, stops, delays, and realistic speech imperfections (corresponding hotkeys [17]) to create a pre-recorded voice cloned message of the *caller*. We went through several iterations with the *caller* up until we produced a pre-recorded message that, in their view, was sufficiently plausible to be indistinguishable to the *callee*. We then downloaded the pre-recorded message and then immediately deleted it from the ElevenLabs repository together with the cloned voice. We resumed the audio recording on the Zoom session and asked the *caller* to call the *callee* and put them on speakerphone. Here, we faced a research design dilemma: to opt for a possible decrease in voice quality and introduction of noise while doing the call on a speakerphone or use a call service like CallFire to upload the pre-recorded message and then place a call to the *callee* that would be routed back on answer to the Zoom session [10]. For the second option, we had to spoof the telephone number of the *caller* through CallFire, obtain the *callee's* number in order to enter it in the call, and upload the *caller's* cloned voice to a third-party server. As we already dealt with sensitive private information (the cloned voice pre-recorded message), we opted not to use the second option and instead used the speakerphone setup. This way, both the *caller* and *callee*'s telephone numbers were unknown to the researchers, and the pre-recorded message stayed only on our computers (subsequently deleted after the interview to preserve anonymity).

Once the *callee* answered the phone (in case they didn't, we tried again a few minutes later, and asked the *caller* to suggest another *callee* that might be able to pick up their call; all *callees* in our study picked up the phone on the first call), the *caller* let them know they are calling them as part of a research study and they are also invited to participate. Then, we, as researchers, provided them with the details of the study and explained that we would like to invite them to listen to a pre-recorded message and answer some questions. After we obtained consent from the *callee*, we played the pre-recorded message of the *caller*'s cloned voice with the choice of pretext. As the cloned voice alone might be distressing, we immediately asked a question about the authenticity of the voice and debriefed the *callee* about the nature of the study to eliminate any possible harm. Here, we offered both the *caller* and the *callee* support if needed and offered them to withdraw from participating in the study (none of the *callees* nor *callers* did so). We then proceeded with the remaining of the questions in our interview (see Appendix B).

4.3 Ethical Deliberation

Even with consenting *caller* and *callee* to participate in our research, and a non-full disclosure protocol approved by our IRB, the subject of our study and the context of *family emergency schemes* and *AI-enabled voice cloning* telephone scams is sensitive and warrants a deliberation relative to both the Belmont and Menlo reports for involving participants in (cybersecurity) research [13, 66]. Relative to the *Respect for Persons* principle, we used an extensive debriefing for both the *caller* and the *callee* to mitigate a potential violation of the participants' autonomy. As part of the debriefing, we offered justification about the need for exposure of a familiar cloned voice for us to obtain results under realistic conditions (*callee*) and a voluntary cloning voice in the first place (*caller*). We offered further resources about telephone scams, AI-enabled voice cloning, and family emergency schemes regularly updated by the Federal Communications Commission [20, 25, 63] as well as the steps on "How to Avoid a Scam" [23], which both the *callee* and the *caller* found

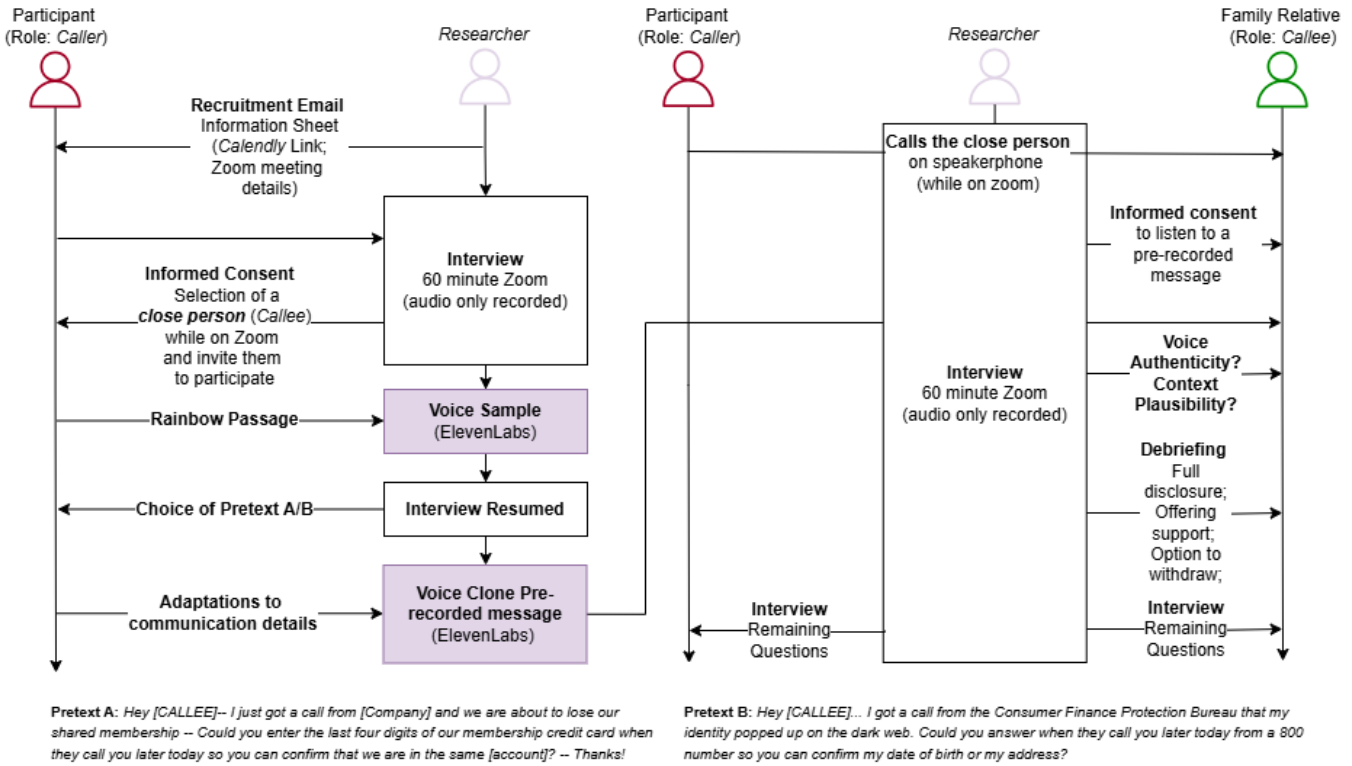


Figure 1: Voice Cloned Family Request – Study Procedure (Pretexts also Listed in Appendix F)

useful. We also offered the option for participants to enter their numbers in the national "Do Not Call" registry list [21] and helped them describe the process of reporting robocalls to FTC.

Relative to the *Beneficence* principle from the Belmont Report [66], we acknowledged that the "burden" of listening to a pre-recorded message was needed to test the *unbiased* utility of the deceptiveness (susceptibility of the *callee* to accept the call as real and hypnotically proceed as asked in the pretext) of the AI-enabled voice cloning in the context of familiar voices that are increasingly used for scam purposes [63]. If we had invited the *callee* on Zoom from the beginning and offered them to listen only to the cloned voice, their attitudes or behaviors would have been *biased* by the fact that they had to hear the pre-recorded message outside of a natural call flow (a condition not present when a real scam call is received). Even if we called the *callee* while on the Zoom session, they would have already known the nature of the call (fully disclosed in the process of obtaining the consent) and acted differently than they would in the normal course of their life where there are distractions and their focus is not entirely set on expecting a call, even if it comes from a known and trusted number.

The *unbiased* utility of the pre-recorded message's impression is a "benefit" that would be enjoyed greatly among individuals targeted by telephone scams, particularly those who might be scammed by familiar voices cloned from audio publicly available on social media or any online repository. The benefits also pertain to any realistic calls that do not even spoof the *caller*'s number or any other type of scam that uses AI-enabled voice cloning. Overall, the benefits of learning the degree of deceptiveness of such calls in realistic settings is a "benefit" that helps articulate the harms associated with voice cloning [33] towards developing user-centered solutions such as contextual warnings that aim to warn users about incoming scam calls [59].

Relative to the *Justice* principle, besides the procedural approval of our study protocol by the IRB of our institution, we offer evidence that the "benefits" of testing the AI-enabled voice cloning in the context of familiar voice scams are equally enjoyed by both the *callee* that bore the "burdens" of listening and the *caller* that bore the "burdens" of consensually volunteering their voice for cloning, that is, both of them were targeted with telephone scams in the past and most probably might be targeted in future. Our sampling criteria strictly stipulated anonymous participation, and we did not recruit participants by any other demographic criterion that might have resulted in social, racial, sexual, and cultural biases (same hold for both the *callee* and the *caller*).

Relative to *Respect for Law and Public Interest* principles from the Menlo Report [13], we borrowed a pretext that was freely available on the FTC's website [25] (*compliance*). We transparently communicated the research purpose with our participants and situated our study in the broader context of harms associated with AI-enabled telephone scams [33], pointing out the need for first-hand experiences in controlled conditions in order to be able to understand the response to this new type of highly deceptive threat. The research results will be used to advocate for contextual warnings [59], call validations [16], regulatory policy updates [64], or any other

user-centered interventions that aim to prevent individuals suffer impersonation harm, economic harm, and harms of mind. During the debriefing, we communicated the risk of harms of mind, in as much as brief and controlled settings, due to exposure of a pre-recorded message of familiar voice cloned and offered the option for both the *caller* and the *callee* to revoke their participation while still being compensated, without any repercussions (none did) (*transparency*). We offer our artifacts, datasets, methodology, and all research instruments for the purpose of replication but also future research, policy-making, or public knowledge (*accountability*).

4.4 Data Collection

We based our semi-structured interviews on our interview script (see Appendix B). To collect data, we arranged audio-only recorded Zoom interviews with initially the *caller* present, and subsequently the *callee* too, on a rolling basis that lasted, on average, 60 minutes. We opted for an audio-only Zoom interview instead of an in-person participation, as online participation as such allowed both the *caller* and the *callee* to answer our questions from the safety and intimacy of their own homes or whether they were at the moment, which we regarded especially important when studying cloned voice pretexts through telephone calls. Initially, the interview transcripts were not anonymized, but we removed any names and references to both the *caller* and the *callee* and deleted the audio recordings altogether (storing them on a secured server). At the end of each Zoom session, we allowed both the *caller* and the *callee* to ask questions about our research, AI-enabled, or any type of telephone scams in general. We also clarified any misconceptions they might have had about telephone scams and voice cloning. The resultant sample demographics are given in Table 1.

Table 1: Demographic Distribution

	Gender	Age	Race/Ethnicity	Education
Pair 1 - **Siblings**	Male *caller*	44	White	Graduate
	Male *callee*	42	White	College
Pair 2 - **Spouses**	Male *caller*	34	White	College
	Female *callee*	36	White	College
Pair 3 - **Siblings**	Female *caller*	72	White	College
	Female *callee*	71	White	College
Pair 4 - **Spouses**	Female *caller*	48	Hispanic	College
	Female *callee*	47	White	College
Pair 5 - **Spouses**	Male *caller*	51	White	College
	Female *callee*	50	White	College
Pair 6 - **Child/Parent**	Female *caller*	48	White	College
	Female *callee*	68	White	College
Pair 7 - **Friends**	Female *caller*	68	Black	College
	Female *callee*	75	Black	College

4.5 Data Analysis

We followed the steps of the practical guide for doing thematic analysis outlined in [8]. The decision to do thematic analysis was part of the research design process where we, collectively as a research team, felt that the degree of subjective response to an impersonated familiar voice and the anonymity would allow us to collect quality, insightful, and relevant information from both the *caller* and the *callee* with sufficient information power [53]. Each of the research members situated themselves both as *insiders* (in the sense that we are target to telephone scams and we encounter AI-enabled voice cloning both through our research and in cyberspace) and *outsiders* (in the sense that we are not familiar with the particular voice or interpersonal nature of the relationship the *caller* and the *callee*). We all felt quite comfortable disengaging in our responses to AI-enabled voice cloning and all types of telephone scams, including family emergency ones, to push the responses to our questions and voice/pretext authenticity/plausibility assessment into the analytical foreground and interrogating them.

In the familiarization phase of our analysis, we immersed and critically engaged with the data. This helped us to approach the coding and develop both semantic and latent codes in relationship to the data (Appendix C). Two researchers engaged in an inductive coding process through multiple rounds for the purpose of collaboratively gaining richer or more nuanced insights (instead of reaching an agreement about every code). We then shifted our focus from codes to themes in order to develop the patterns or meaning across our dataset and cluster the codes around central organizing concepts about inauthentic content. We first generated initial themes, then, working through a thematic mapping, we developed and reviewed the resultant themes. The continuous analytical refinement enables us to define and name the themes. Lastly, we wrote our results around the themes, selecting data extracts to evidence our claims and allow one to independently judge the fit between the data and our understanding and interpretation of them [8].

4.6 Positionality Statement

We recognize that our identities and our background influence our research and subjectively bias our thematic analysis [8]. Our active role as researchers in the process of knowledge production shapes the interpretation of the results relative to our particular social, cultural, historical, disciplinary, political, and ideological positionings. Our research team is composed of one cisgender woman and three cisgender men coming from universities in the United States and Germany. Two researchers are in their 30s, and two researchers are in their 40s. All researchers have an educational background in computer science. All researchers have experience with various types of telephone scams, email, SMS, and any other types of general social engineering encountered in cyberspace. All researchers have at least at least 36 months of experience with AI-enabled voice cloning. None of the researchers are involved in the development of any AI tools that enable voice cloning, nor any tools that facilitate telephone scams, automatic dialing, or any form of robocalling. Our motivation for exploring AI-enabled voice cloning scams with familiar voices and using known numbers (to the *callee*, not us researchers) stem from our past lived experiences and experiences studying telephone scams and observations of how various forms

of deception, including using a synthesized media, could shape user behavior, perceptions, and decision making in cyberspace. With this study, we aimed to encourage endeavors towards exploring this new type of emergent scam that could induce multiple harms, not least harms of mind.

5 Findings

This section presents the final themes we agreed upon after engaging in a reflexive thematic analysis process aimed at addressing the following research questions:

- **RQ1**: How do individuals that receive phone calls from a known telephone number (Caller ID) and a familiar voice (cloned through AI) respond to the (a) voice authenticity; and (b) plausibility of the call's context?
- **RQ2**: How AI-enabled voice cloning could be used to target individuals, in both telephone scams and general deception?
- **RQ3**: What protections should exist against AI-enabled voice cloning telephone scams using familiar voices?

5.1 Theme I: Creepy Feeling

The first theme that we developed out of the *callee*'s response was associated with the induced feeling of "creepiness." Scientifically defined as the "anxiety aroused by the ambiguity of whether there is something to fear or not and/or by the ambiguity of the precise nature of the threat (e.g., loss of membership or identity theft) might be present" [44], the creepiness is a state of uncertainty that results in a pause as to how one should respond. Consequently, the feeling of being "creeped out" is a state of reservation because it would be considered rude and embarrassing to run away from a person (a familiar-sounding *caller*, in our case) who has done nothing overtly threatening (a non-urgent request for sharing last four credit card digits or answering a future call, per our pretexts), but, on the other hand, it could be perilous to ignore one's intuition (is the voice authentic? is the context of the call plausible?) and remain in a potentially dangerous interaction (being scammed).

All the *callees* selected by the *caller* in our sample answered the call. They indicated that they see no reason not to pick up again in a hypothetical scenario where the voice on the other end starts speaking right away. Regardless of the use of speakerphone (a slightly compromised quality) and the fact they "expected" to listen to a call instead of being cold called (as in a real telephone scam), the *callees* faced ambiguity about the voice because it sounded like the *caller*, though something was off. Here, **P2*callee*** explained:

> "*The voice was really, really, really close. Other than the cadence of the speech, the [caller's] voice was theirs, for sure. But, it was creepy. It sounded just like [the caller], only a bit more faster than usual.*"

As to the ambiguity of the nature of the threat, **P2** added that "*the message sounded as a request, but in this case, it made it think whether I need to urgently do something or not.*" The cloned voice of the *caller* did induce a bit of uncertainty, but **P4*callee*** found the context of the call plausible and convincing nonetheless:

> "*The context of the call was really real. [The caller] is always calling me for a password and that kind of stuff, so I didn't really think much of that. [The caller] was literally telling me to respond to something, and I actually was about to get my phone to see if I got some kind of text of something that I had to respond to. I've always got my headphones on, so [The caller] will just say hello to get my attention, so I check my watch to see what message [the caller] sent me. This time I was busy in a different room, and because I had my headphones on, I thought [the caller] was trying to get my attention to help with the Netflix account. I was just about to press zero and go get the information. I was a little hesitant, but I was still about to do it.*"

Some of the *callees* expressed reservations due to the quality of the pre-recorded message and the nature of the voice that indeed sounded familiar and came from the *caller's* number, but it "*sounded very robotic and [the caller's] tone wasn't changing at all*" [**P5**$_{callee}$]. However, the *callees* decided to remain in interaction despite the ambiguities. Closely resembling the *family emergency scheme* of the AI-enabled telephone scams [63], **P6**$_{callee}$, overrode any ambiguity about the quality and nature of the voice and followed suit as requested by the pre-recorded message we played to them:

> "*I thought it was real sounding. I pressed zero like it asked. It was my daughter's voice, and she needed me to do something.*"

Participant **P1**$_{callee}$, too, captured the combination of voice authenticity, trusted Caller ID, and plausibility of the call context as fully persuasive and working:

> "*You creep! It really worked! I had it on speakerphone because I saw it was my brother. Immediately, I thought, oh, that's [the caller]. I thought I was going to have to call mom and warn her.*"

5.2 Theme II: Family Emergency Scheme

Though we deliberately excluded the "emergency" aspect from the *family emergency schemes* [63] due to a potentially non-minimal risk of harms to mind to the *callees* and possibly the *callers*, the cloning of a familiar voice was nonetheless seen as most powerful in targeting individuals in the context of emergency. Participant **P6**$_{callee}$, in no uncertain terms, pointed that:

> "*I would do anything for my daughter, especially when I hear her voice on the other end. I would have done what it asked. I would give a scam call money if I thought it might be for any of my children without asking.*"

Participant **P5**$_{callee}$ further expanded:

> "*I've gotten calls about insurance, the IRS, and those scams. But I don't respond because I honestly see it's a scam right away. But this one, it was just it was very, very creepy as [the caller] sounded like they had a gun to his head the way he was talking. The call felt like [the caller] wasn't even talking to me. So it might well be some emergency that I don't know off and can't guess at the moment, but I have to respond appropriately.*"

Here, we asked the *callers*, first, if and how realistic it is for a telephone scammer to obtain their voice to create a sample as the initial prerequisite for the cloning part. Participants agreed that this is quite easy and that there are several ways to do it. For example,

P4$_{caller}$ pointed to an option where the telephone scammer could "*give them a first, and if [they] were to talk to them then they could get a good voice sample and use it to scam another person they could figure out I am close, searching online or social media.*" Participants **P5**$_{caller}$ and **P6**$_{caller}$ noted that they "*teach courses and have recorded lectures available on YouTube, so anyone could get [their] voice from one of the lectures*," and pointed that "*students or anyone could also record [them] in class too.*" Participant **P7**$_{caller}$ invoked the very recording of interviews part of research studies where "*the scammer could well pose as a researcher, ask me bogus questions to get a good sample of my voice, and instead of the answers, use it to clone it.*" (we ensured the *caller* we are not scammers and that they could verify our credentials on our universities' websites or Google Scholar).

Following the scheme, we next asked the *callers* why and how one would decide to impersonate them for the purpose of a telephone scam. Participant **P1**$_{caller}$ saw a financial fraud opportunity that targets "*known associates, friends, or anyone familiar with [their] voice from podcasts and gaming events where [they] speak and target them with some sort of promotion or exclusive access.*" Participant **P6**$_{caller}$ saw an identity theft opportunity where "*someone could use [their] voice to call someone at [their] place of work as an authentication and get sensitive information*" (interestingly, such incidents happened in the real world when a voice cloned message of the LastPass CEO was left on the phone of a LastPass employee, asking for documents [37] and a voice cloned message of a UK-based company CEO persuaded an employee to transfer almost a quarter of a million pounds to a scammer controlled account [11]). And Participant **P3**$_{caller}$ saw an opportunity for enhancing the chances of already used and tried pretexts for both financial fraud and identity theft "*by targeting their social security information.*"

Both the *callers* and the *callees* saw various scenarios where a familiar voice could be used for the purpose of a scam. For example, **P6**$_{callee}$ noted that the family emergency scheme could get more advanced and persistent over time as "*scammers could start with vague, less concerning requests to engage with the target over several calls, learn their response, and then create a realistic scenario knowing the callee will certainly respond to their request*," varying the spoofed Caller ID as needed. Both the **P1**$_{caller}$ and **P1**$_{callee}$ felt that the AI-enabled voice cloning, if not for family or relatives voice, would certainly work and will be used in future for "*targeting people with voices of politicians or celebrities, be that for political campaigns or scamming money out of them.*" Anecdotally, **P7**$_{callee}$ mentioned an option where voice cloning could be well used in conjunction with generative multimedia AI for a family emergency scheme:

> "*My son just did this to me! He called me with a fake voice. I knew it was fake, so I hung up on him and called him back to yell at him! He knows I like AI research. He thought I'd think it was fun. It was interesting. But what if I get a fake voice call and a fake message with AI-generated injury or someone kept in captivity*"
> (Note: The son was not in our study, nor were any AI-generated images involved during the interview)

5.3 Theme III: Family/Friend Code Word

Given the first-hand experience with the familiar cloned voice telephone scams as part of the study, we asked both the *caller* and the *callee* what, in their view, could be the way forward in countering such deceptive schemes with a high probability of causing multiple harms to individuals. Most of the participants expressed skepticism about any regulatory protections because, as **P1**$_{caller}$ put it, "*criminalization is a start, but that's reactionary; Not sure what could be done to prevent it proactively.*" Seeing a rather limited option for proactive protection, participants reflected that a technical solution might be needed. Here, **P3**$_{caller}$ thought that "*maybe [the telephone providers] could warn you at the beginning of the call that this call may be fake*" referring to the STIR/SHAKEN possibility for a telephone provider to insert a call warning such as "Scam Likely" [16]. Past work showed that this verification could be easily circumvented by spoofing the Caller ID [56], so recent work has proposed using contextual warnings that are based not on the Caller ID but on the call context instead so *callees* are informed about the nature of a scam, for example a banking one or identity theft one [59] (Google, for now, is testing this option for Pixel and Google Fi users, powered by their AI engine, Gemini [31]).

But these technical solutions might take time to take effect and, as participants expressed, the "*scammers are getting better with AI much faster than technology can cope with*" [**P7**$_{callee}$]. A common theme, thus, emerged among participants relative to how family members or friends could protect themselves using a *family code word or friend password*, a specific piece of shared information known only to family members, relatives, or close friends. Participant [**P5**$_{callee}$], here, suggested:

> "*With AI, being aware that cloning a voice you are familiar with would inevitably happen. That's why we've talked among us in the family about code words and things like that, so we're asking very personal questions if we feel something is amiss on a call or a message. [Scammers] can clone the voice, but I see no actual way for [the scammers] to clone the actual answers or flow of answers to these personal questions.*"

Their spouse, [**P5**$_{caller}$], added:

> "*We've discussed having family code words in the past, and that works in the immediate circle. But when we talk about code words to the older folks in our family, we have to be super careful in choosing one. They will for sure suggest a code word to be their address. And you have to explain that any scammer can find their address online, so that won't work. Next, they will suggest asking about their age. That's available online, too. And so we try to explain, you know, the best code word would be just something like, you know, "supervisor," something that has nothing to do with anything. So I think code words are good, but it's not easy implementing them for the extended family.*"

This within-family protection was especially needed for older adults, as, in the view of our participants, seniors are especially vulnerable categories of AI-enabled voice-cloning telephone scams. Educational efforts or demonstrations that reveal the deceptive nature of telephone scams may be effective in countering scammers' advantages. Participant [**P4**$_{caller}$] shared:

> "*I can see my mom falling through something like this where my voice is cloned over the phone. You gotta make*

people aware of the threat, you know. I'm gonna lecture my in-laws and my parents after this experience. Don't listen to phone calls mindlessly, because it does sound so real, especially for older people, but think to ask me something specific about me or our family. Being able to do this to show them how real it can be is a good learning tool, that's for sure."

6 Discussion

6.1 Trust Implications

Our results bring several trust implications to the fore: (i) the authenticity of the voice on the other end; (ii) the ability to protect one's own identity, i.e., their voice, from unauthorized misuse; and (iii) the telephone providers and regulatory bodies role to prevent AI-enabled voice cloning scams. That our *callees* felt creepy is not anecdotal evidence of the unusually deceptive nature of the voice cloning, but it well aligns with the "*uncanny valley effect*" or the switch between the expected empathy (or simple ambivalence) towards the *caller* to an unexpected creepiness when faced with they perceive as potentially sudden an unusual request [47]. The loss of trust in the voice on the other end, effectuated by questioning the realness and credibility of the *caller* as a persona, might result in so-called "*doppelganger-phobia*" or negative emotional reactions and skepticism caused by the abusive potential of AI clones to exploit and displace the identity of a familiar individual [38].

It's not just the trust setback that *callees* experience as a result of the familiar voice clone/number scam. The cloning of someone's voice, if done time and again and over several scams, threatens the *callers* with "*identity fragmentation*," that without any way for the *callers* to control, undermines their societal right for cohesive self-perception and unique individuality (recall, for example, the incident with the OpenAI's creation of a voice that was "eerily similar" to Scarlett Johansson's, despite her declining to contribute her voice to the platform, only worse, where the cloning entity is not a company but scammers) [58]. This inability to reign in the AI-enabled voice cloning, in turn, might result in skepticism in what hereto was assumed as a trustful infrastructure for telephone, messaging, and internet communication.

Even if one could screen calls and use protections such as Scam Block or Scam ID that current telephone operators (at least in the US) offer, these protections are based on inferences where a scammer targets a large set of *callees* with numbers that are unfamiliar to them (even if they use what's called "*neighbor spoofing*" where the Caller ID comes from a local area code where the *callee* resides or matches the area code of their number), and the "robocall" is a pre-recorded message to which many users are accustomed too. The scheme we tested does the opposite of this strategy – it uses a full Caller ID spoofing (and even if it doesn't and resorts to neighbor spoofing), it targets one, instead of a range of *callees*, and clones a voice of a familiar *caller*. Scam Blocks, Scam ID, or the likes [26]) have no reason to stop such a call as they seek unusual call patterns, not looking for a pre-recorded message that might be a familiar voice cloned. Even if call traceback is enabled to trace the scammers [1], nothing precludes them from posing as legal telemarketers or polling entities and instantiate the AI-enabled voice clone scam and avoid tracing.

All of these trust implications, along with the overarching perceived loss in the way people communicate with each other in real-time, reveal a situation where the dual use of voice cloning technology is promulgated without accountability or responsibility of the potential adverse consequences [74]. Without an intervention (legal, regulatory), platforms are hardly expected to take any responsibility for preventing unauthorized AI generation. One needs not to go far back to recall the (social media) platforms' involvement of disinformation promulgation and their successful blame "shift," reasserting that the primary responsibility lies with the users and those who create false content, not the platforms who are merely providing the infrastructure [57]. AI content, in general, falls under the definition of misinformation and when promulgated with intention such as in the case of voice cloning, does create the same predicament for the so-called "enablers" or platforms that offer voice cloning or conversion capabilities. Currently, a clone of someone's voice can be created with only a check of a box that indicates consent to use the voice, but that doesn't give a full grasp of the potential negative externalities in doing so. Voice cloning/conversion platforms, thus, are not currently held accountable or expected to implement solutions to prevent their unauthorized use [73], even though they do plan a more involved role what was an entirely separate entity in the case of disinformation creation and dissemination – "those who create false content."

6.2 Regulatory Implications

Unwanted or scam calls are illegal, and multiple statutes (a list of US statutes can be found here [27]) and the Federal Communication Commission (FCC) have made AI-generated voices in robocalls illegal under the Telephone Consumer Protection Act (TCPA) [20]. While these steps are important in curbing the threat of targeted familiar voice cloning telephone scam schemes, there is an absence of a coordinated legal approach targeting regulation of the ecosystem of actors involved in this or similar deceptive contexts [36]. Equally important is the focus on individuals who are victims of the familiar voice clones created without their consent and knowledge. Here, the states of California, Illinois, Texas, and Washington have laws that require explicit consent before the collection or use of any biometric data, including voice data, with the intent of preventing unauthorized voice cloning and subsequent misuse [39]. Additionally, Tennessee has enacted the Ensuring Likeness, Voice, and Image Security (ELVIS) Act, which grants individuals the right to control the commercial use of their voice and assigns authority over a person's voice for ten years following their death [35].

In the absence of a federal law and a defined ecosystem regulation, individual states are beginning to address the gaps in existing legislation that hinder the prosecution of AI-related offenses. In 2024 alone, 45 states introduced AI-related bills, which varied greatly in their regulation over a wide variety of AI technology, but only 31 succeeded in passing them. Efforts to establish protections for voice cloning in states like Alaska, Arizona, Florida, and New Jersey either failed or remain pending [51]. The remaining states have yet to propose or adopt laws that would protect individuals from unauthorized voice cloning. This regulatory shortfall, taken together with the trust implications stemming from our study, does little to signal that individuals, in the role of *callers*, have clear legal

recourse against voice cloning. The same could be said about individuals in the role of *callees*, as their options are reporting a scam or putting their phone number on FTC's Do-Not-Call Registry [24], but this helps little to save themselves from being targeted with a family emergency or any of the traditional scam schemes [22].

6.3 Usable Security Interventions

Non-technical solutions currently exist, such as the shared family/friend code word [4]. Education on AI generated voice cloning scams and *family emergency schemes* would allow individuals to enact these methods. The human detection of a potential AI-generated voice provides the *callee* with the opportunity to call the user or another trusted family member back to confirm the validity of the call. Unfortunately, these solutions rely on the ability of people to detect that the call is AI-generated and the creation of strong code words. Ideally, a technical solution would be integrated into a system and be able to intervene, assisting the *callee* in determining the correct course of action in an emotionally driven situation.

Past research shows that people screen spam calls either by the Caller ID [67], by the presence of a STIR/SHAKEN indicator (e.g., "Scam Likely") in addition to the Caller ID [16], or by sending it to "Voice Mail" and listening to the message/reading the transcript without the pressure to respond to it immediately [59]. But the Caller ID and STIR/SHAKEN indicators as cues won't work if the call comes from a spoofed, yet *trusted* number that the target has in their contacts and regularly receives calls from. The screening of the "Voice Mail" (or its transcript) might still help, but that is predicated that the scammer will actually leave a Voice Mail in the first place. If that happens, individuals are left on their own faculties to decide if the message is plausible and the voice, if cloned, is authentic and matches the identity of the calling number.

For traditional scams that do not use voice cloning, one proposed user-centered intervention was an implementation of a virtual assistant that interacts with the scammer on behalf of the user and only lets the call go through if no attempt for defraud/identity theft was detected [54]. But such a solution might be dismissed by the *callee* simply because the call is from a *trusted* person and does not need to be screened by an intermediary (allow-listed). Call blocking or "anti-robocall" apps, here, for the same reason, will not work as their goal is to block or filter out *non-trusted* numbers, not numbers of the immediate family of friends [14]. A promising solution is to alert the *callees* about the context of the request of an incoming call that is suspected to be a scam. Google demonstrated a proof of concept as a built-in Android OS feature that uses Gemini Nano (Google's generative AI engine) to dynamically offer assistance on phone screens immediately after a call is answered [31]. The use of a generative AI engine requires the *callee* to enable the feature. *Callees* in prior research have presented security concerns about allowing generative AI to access their calls [2, 59]. The difference with this alert, shown in Figure 2, is that it is not just alerting the user that the call is "likely a scam," but it explains what kind of scam it is (in this case, the context "*Banks will never ask you to move your money to keep safe.*" refers to investment, money-making or similar financial fraud schemes [25]).

The warning is not associated with the Caller ID, giving the opportunity to be customized for family emergency schemes or any AI-enabled voice cloning, and it offers the option for a *callee* to proceed with the call if they believe the call is in fact legitimate. While this was just a proof of concept, a similar idea – only aural warnings, instead of a visual screen warning – were evaluated for context provisioning relative to imposter and investment/money-making scams [59]. The aural warnings, shown in Table 2, were created by generic voices akin to Siri, Cortana, and other known voice assistants on phones, with the idea to warn the *callee* on the go so that the benefit of the alert is enjoyed not just by people who can see the phone screen at the time of the call, but also by people with visual disabilities and in situations when phones are answered on earphones, in cars, or hands-free.

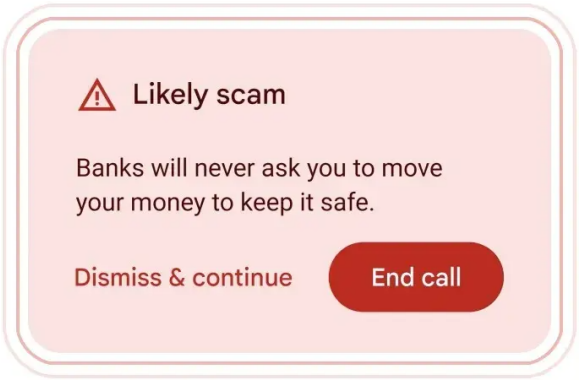


Figure 2: Contextual Telephone Scam Warning

Table 2: Aural Contextual Scam Warnings

Scam Scheme	Contextual Aural Warning
Imposter	`This call is likely a scam. Government agencies will never ask about your Social Security number or discuss legal matters over the phone. Hang up immediately, otherwise, proceed with caution.`
Investment	`This call is likely a scam. Banks will never time pressure you to make any change to your account over the phone. Hang up immediately, otherwise, proceed with caution.`

These two user-centered solutions might offer an opportunity to alert the *callee* about incoming family emergency schemes that users a familiar voice cloned. As an input on the detection of a voice cloned scam, the FTC has recently run a "*Voice Cloning Challenge*" [64] where real-time solutions offer sampling and screening of a call (on the privacy challenges and feasibility of baseline comparison with all of these issues see Appendix E) to determine discrepancies and the level of liveliness of the speech to detect whether a voice is cloned or not. Assuming this is possible (for one, there is a consent and a baseline sample of the *caller's* voice for comparison), we envision adaptations of the contextual warnings akin to the ones we show in Figure 3 and Table 3, based on the framework for labeling AI-generated content proposed in [75].

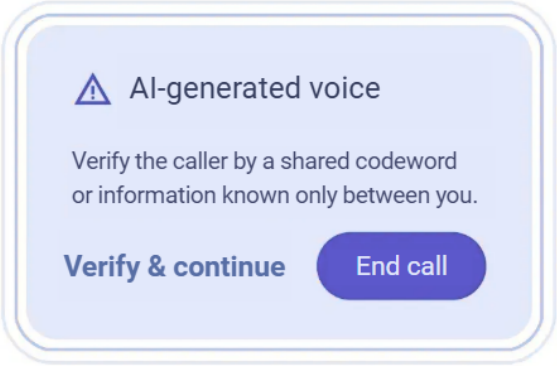


Figure 3: Familiar Voice Cloning Warning

The context is not *per se* emphasized in the warning (as it might be too subjective for the *callee* and unlike in the examples in Table 2 where it could be inferred by the device or telephone operator when the scam targets a wide range of *callees* as its usually the case), but instead urged the authenticate, per our findings, with a family code word. This way, we believe the intrusion is minimal and allows for both the *callee* and the *caller* to retain their agency and autonomy in deciding how to proceed with the call and eventual reporting of possible voice impersonation to FTC.

Table 3: Familiar Voice Cloning Warning

Scam Scheme	Contextual Aural Warning
Familiar voice	`The voice on this call is likely generated with AI. Verify the caller by a shared code word or information known only between you, otherwise, proceed with caution.`

Additionally, the integration of a *callback* feature, shown in Figure 4, would allow the user to directly contact their family/relative's number from their contacts rather than engaging with a call originating from a potentially spoofed Caller ID. The flow of a call with the callback feature is shown in Figure 5. When an incoming call is received, first a brief sampling and screening of the audio is performed against a saved voice sample, capturing a short segment of the caller's voice for real-time analysis. If it is determined the speech may be a voice clone, a warning message would immediately appear on the user's phone before they accept the call, notifying them that the call may involve a cloned or artificially generated voice. At this point, the user would be given the option to use the integrated *callback* feature to safely contact the verified number stored in their contacts. For example, if a mother were to receive a call from a potentially spoofed Caller ID showing her child's number (as in a real-world incident involving a "kidnapped" daughter [65]), the system would detect anomalies in the caller's voice, issue a warning message, and offer her the option to immediately place a verified callback to her child's stored number, thereby avoiding engagement with the potentially fraudulent caller.



Figure 4: Callback Feature With Familiar Voice Clone

6.4 Limitations

The realistic settings impose several limitations pertaining to our study. The sample size and the demographic distribution limit the generalization of the results to the entire population of individuals targeted with general or *family emergency scheme* scams. The recruitment stipulates an establishment and maintenance of a continuous trust between the participants pairs' and the researchers and precludes expansion of large samples for testing, though this is anyhow a future goal of our work. We selected the US as our initial place of investigation based on the reports suggesting that Americans on average get 16 scam calls a month, but we nonetheless plan to continue this line of investigation with a larger sample that spans beyond US and captures *family emergency schemes* facilitated by *voice cloning* across the globe (for example, Brazilians receive even 26 scam calls per month and Indonesians, as well as Spaniards, 16 scams call per month too [32]). The composition of our sample is skewed towards participants who have higher education, majority women, and experienced adults (Table 1), so the future expansion of the sample will also balance along these demographic lines.

A limitation comes from the fact that we did not cold call the *callee* to obtain the actual response on picking up the phone and answering the pre-recorded message, a call flow fully resembling an actual scam instance [63]. The choice of pretexts is also a limiting factor because the text might not fully resemble the actual, expected, conversational specifics that the *callee* and the *caller* are used to. A limiting factor is our choice to present the pre-recorded message over a speakerphone and not directly through an actual telephone channel (which we judged as a more than minimal risk to privacy).

ElevenLabs, as the AI voice cloning engine of choice, is also a limitation as other voice synthesizers might produce pre-recorded messages of quality other than the ones we had (we did check about the convincing quality and linguistic elements of the pre-recorded message with the *caller* before we played it to the *callee*, but this was a subjective determination on their side). We are limited to the fact that we used the English language for the test of the scam in the native accents of the *callers*. Other languages and other

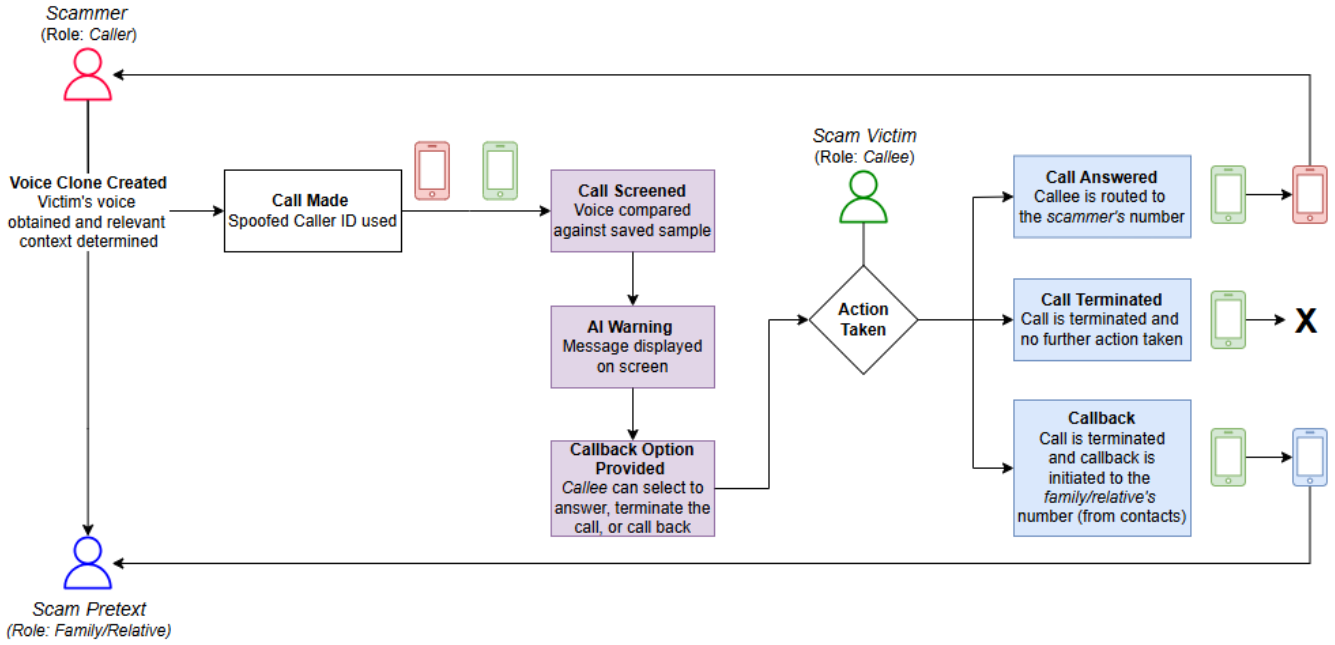


Figure 5: Voice Cloned Family Request – Flow of Call with Callback Feature

accents might yield different results from ours. Though we left the *callee* sufficient time and support to engage with the pre-recorded message, this might not have been sufficient for them to formulate a more informed expression about their overall assessment and ultimate decision about it.

7 Conclusion

To learn first-hand experiences with AI-enabled voice cloning of familiar voices used as part of telephone scam schemes, we interviewed 14 participants (7 pairs of *callers* and *callees*) who volunteered to have their voice cloned (*caller's voice*), being called from a trusted Caller ID (*caller's number*), and listen to a pre-recorded message (*callee*). Through a thematic analysis, we found that the cloned voice, in conjunction with the trusted Caller ID, gave participants pause as it sounded authentic and made them think about the plausibility of the call context. As this impression, if invoked through a real scam call, could sufficiently persuade the *callee* to follow through with the scammer's request (we used two pretext requests about shared membership and identity theft), participants saw a possible defense in using a shared family or friend code word. Based on our findings and building on past work on user-centered scam alerting and AI-generated content labeling, we proposed contextual warnings in both visual and aural variants. AI-enabled voice cloning or any other type of impersonation, in our view, would not go away, at least for the time being, so we deem it pressingly important to bring the evidence of this highly deceptive practice and inspire responses that aim to protect individuals without the need to compromise, renegotiate, or reestablish the trusting relationships they already have with their family, relatives, or friends.

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A Recruitment Email

From: Researcher's Email
Subject: Research Study Participation
Date:
To: Potential Participant

Hello,

My name is Filipo Sharevski, Ph.D., a professor at DePaul. I am conducting a study to learn more about how people interact with generative voice AI. We are asking you to be part of the research study because you have already explored cloning your voice with a generative voice AI tool, such as ElevenLabs.

If you participate in our research study, you will be compensated with a $60 Amazon Gift Card for 60 minutes of participation ($40 Amazon gift card for the called party as part of the interview, sent via their email provided during the interview – not part of the recording).

If you are interested in participating, please use the following Calendly link to find and schedule a timeslot that best suits your availability. Once you choose a timeslot and confirm in Calendly, you will receive an email with details on how to join the research interview, which will be hosted via Zoom or a phone call.

https://calendly.com/filipo-sharevski/deceptive-voice-content

If you have any questions, concerns, or are unable to use the above Calendly link to schedule your interview, please feel free to reach out directly to me either via email, text, or phone call. You can also respond directly to this email. My email is fsharevs@depaul.edu, and my cell phone number is +1 765-714-9574.

Thank you for your time in reading this email.

B Study Questionnaire

B.1 Questions for *Caller*:

(1) What's your choice of pretext A/B?
(2) Why did you choose the target you selected to be a "*Callee*"?
(3) What is a realistic way someone could clone your voice?
(4) Who might impersonate you?
(5) Who might target the "*Callee*" with an impersonated voice of you?

B.2 Questions for *Callee*:

(1) What were your initial thoughts about the pre-recorded message?
(2) Did the recoding sounded accurate of the "*Caller's*" voice?
(3) Was the content of the correspondent to a plausible scenario (i.e., is it plausible that you might receive such a voice from the "*Caller*")?
(4) Did you suspect, even in a brief moment, that someone might have impersonated or cloned the voice of the "*Caller*"?
(5) What cues did you use to determine the pre-recorded message might not have been left by the real "*Caller*" (i.e., it was fake)?
(6) Assuming you would have picked up, and this was a real call, would (and how) you have proceeded as the voice on the other hand on the phone?

B.3 Questions for Both:

(1) Have they heard of any other calls or have received pre-recorded messages generated by AI voices?
(2) What would they do if a call (or a pre-recorded message) sounds very realistic, but you might suspect that something is amiss with the nature of the call, the request, the time you got it, etc.?
(3) Where might these types of phone impersonations through voice cloning might go in the future (image, audio, video)?
(4) What protections do they think should be in place to warn people about such calls?

B.4 Demographics ("*Caller* and *Callee*"):

(1) Age
(2) Gender
(3) Ethnicity/race

(4) Level of education
(5) Experiences with unwanted calls (e.g., Often, Sometimes, Never)

C Codebook

C.1 Initial Response

- **Reserved** -The *callee* suspected something was amiss in the nature of the voice infliction, tone, and prosody, but the voice was authentic to the one associated with, and the call context was deemed plausible
- **Creepy** - The pre-recorded message sounded creepy in the sense that the voice was authentic to the one associated with and the call context was deemed plausible, but the *callee* was uncertain about whether the request was real or not
- **Confused** - The *callee* suspected something was amiss in the call context, but the voice was deemed authentic to the one associated with and the call

C.2 Real World Scenarios

- **Emergency** - The *caller* and/or *callee* felt the cloned voice would work well for an emergency as it would override the reservations, the feeling of creepiness, and the confusion of either the cloning inconsistencies or implausibility of the call context
- **Targeting Older Adults** - The *caller* and/or *callee* identified "older adults" as the population specifically vulnerable to the familiar cloned voice / trusted Caller ID telephone scams

C.3 Protections

- **Family Code Words** - The *caller* and/or *callee* saw the only protection to these types of scams by using family code words known only to them and hard to be discovered and cloned
- **Personal Questions** - The *caller* and/or *callee* saw a possible exchange of challenge/response questions for elaborate schemes that might use several cloned voice messages as a form of testing a shared secret

D Debriefing

Thank you for participating in our research on how people interact with generative voice AI. This study aimed to examine whether people are able to notice if a conversational recording is created by a conversational large language model or by real interlocutors. So far, no research exists on how users deal with recordings generated using a familiar voice and relevant content. This is why we asked you to provide your feedback.

It was necessary for the researchers to withhold this information from you regarding the purpose of the study to ensure that your actions and answers to questions accurately reflected your cybersecurity hygiene, perceptions, and beliefs. Your participation in the study is important in helping researchers identify the best ways to address the deceptive pretexts in phone calls that are generated using a conversational large language model and seeded with voices from real people you may know or whose voice you are familiar with. Since we did not collect any personal information, we are not able to remove your entry from the data bank of our research interviews.

The final results of this study will be published in a peer-reviewed journal. Your results will not be available individually, and your participation will remain confidential. We do not keep, record, or collect any personal credentials. If you have any additional inquiries, please contact Filipo Sharevski, Ph.D. at fsharevs@cdm.depaul.edu. If you have questions about your rights as a research subject, you may contact Jessica Bloom in the Office of Research Services at +1 312-362-7388 or via email at jbloom8@depaul.edu. You may also contact DePaul's Office of Research Services if your questions, concerns, or complaints are not being answered by the research team, you cannot reach the research team, or you want to talk to someone besides the research team.

E Privacy Challenges for User-Centered Scam Interventions

The user-centered anti-scam solutions presented require *sampling* calls for conversation patterns commonly associated with scams or voice cloning and, using a generative AI engine, dynamically *inserting* the warning before the call is connected. Done through an AI engine, obviously, this approach raises alarms about the peril of privacy intrusions, censorship, or centralized control [12]. It is yet to be seen whether users will opt in for such protection or allow their original voice to be sampled in order for a detection/protection engine to infer possible impersonation of their voice. We expect that individuals would justifiably be reluctant to do so, given the negative experiences in the past with mass telephone surveillance in the US [15] and conjectures based on constantly growing evidence of data breaches [43] and generative AI incidents [3, 5]. A way forward, perhaps, might be to retain the AI approach but offer the option for telephone users to *donate* samples of their voice or choice for the purpose of generating contextual warnings and an act of altruism [71]. We see this as a starting point, aided by privacy guarantees and full transparency of warning generation, implementation, and provision processes.

F Study Pretexts

- **Pretext A**: Hey [CALLEE]– I just got a call from [Company] and we are about to lose our shared membership – Could you enter the last four digits of our membership credit card when they call you later today so you can confirm that we are in the same [account]? – Thanks!

- **Pretext B**: Hey [CALLEE]– I got a call from the Consumer Finance Protection Bureau that my identity popped up on the dark web. Could you answer when they call you later today from a 800 number so you can confirm my date of birth or my address?